

Electronic Products Dataset (EDA)

HNG Internship Data Analytics Track

Stage 0 Report

Emmyy DA

Dataset Overview & Preparation

- **Dataset Overview:**

- The dataset contains over 10,000 product records, and 13 columns including attributes such as product name, brand, category, price, stock levels, size and availability.

- **Data Preparation:**

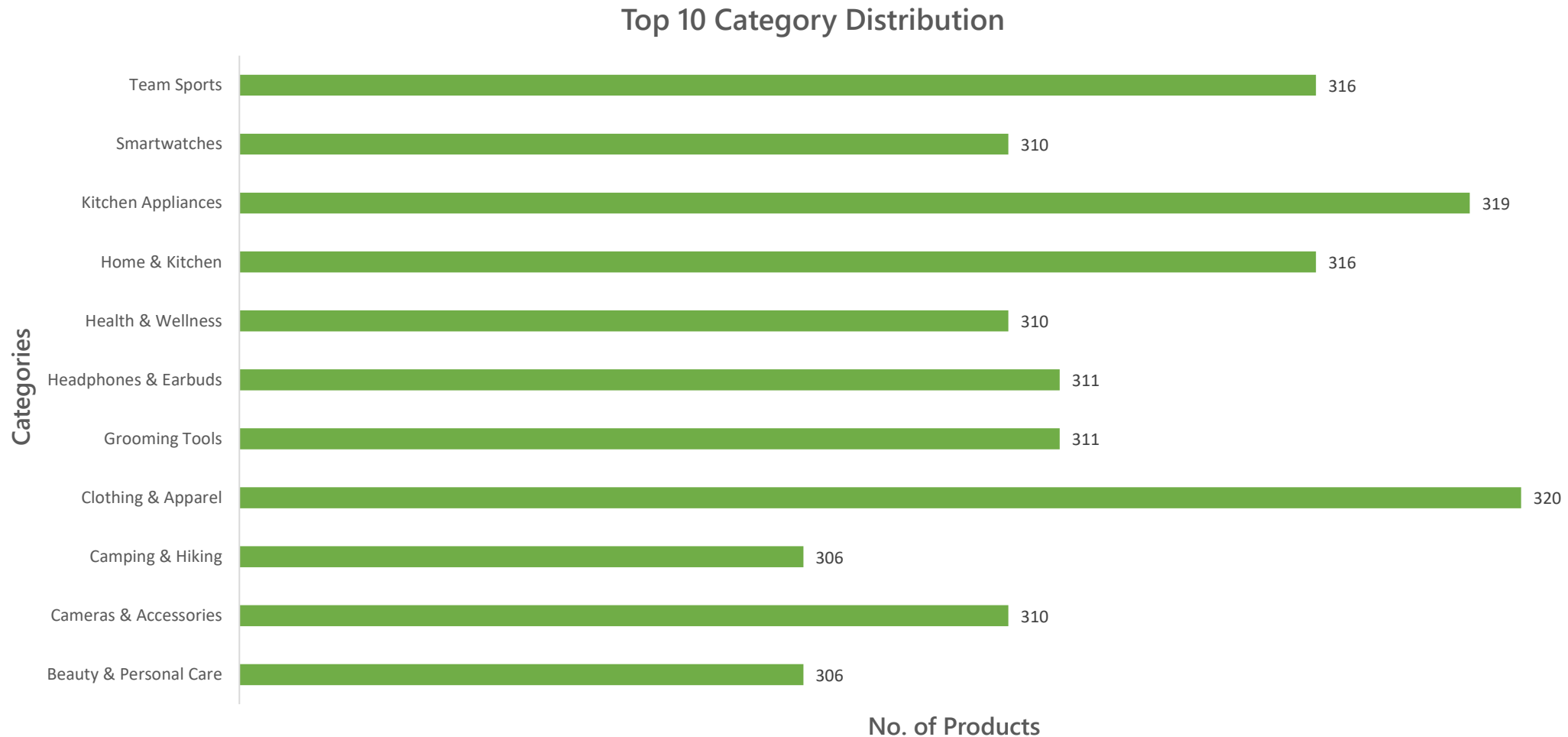
- Converted price values to USD currency format.
- Standardized availability labels for consistency.
- Ensured correct data types across columns.

- **First Observation:**

- The dataset is large and diverse.
- Inconsistent size formats across the datasets.
- The description field has little or no meaningful relationship to the dataset.
- The dataset is not entirely focused on electronics, but includes other product categories.

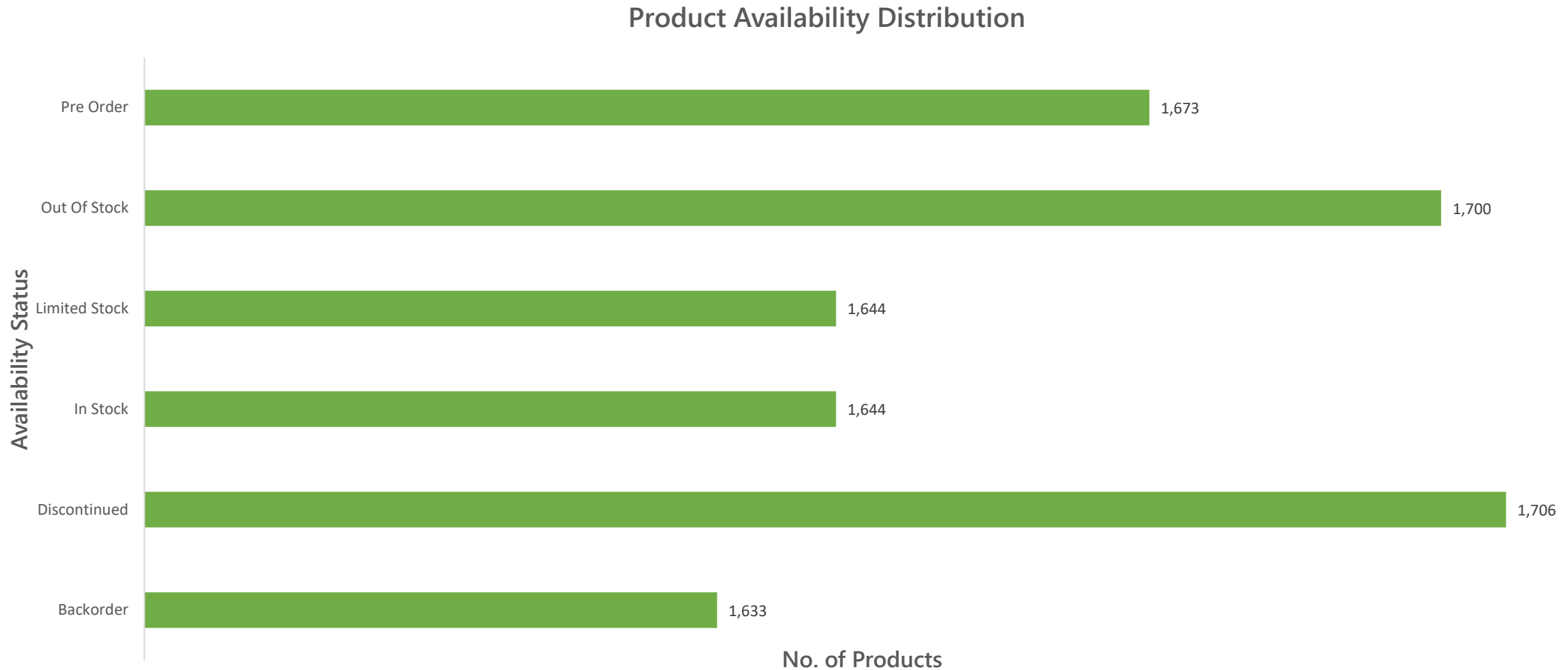
Q1. How are the products distributed across different categories?

Product distribution is evenly spread across categories, with Clothing & Apparel having a higher count, indicating a balanced dataset.



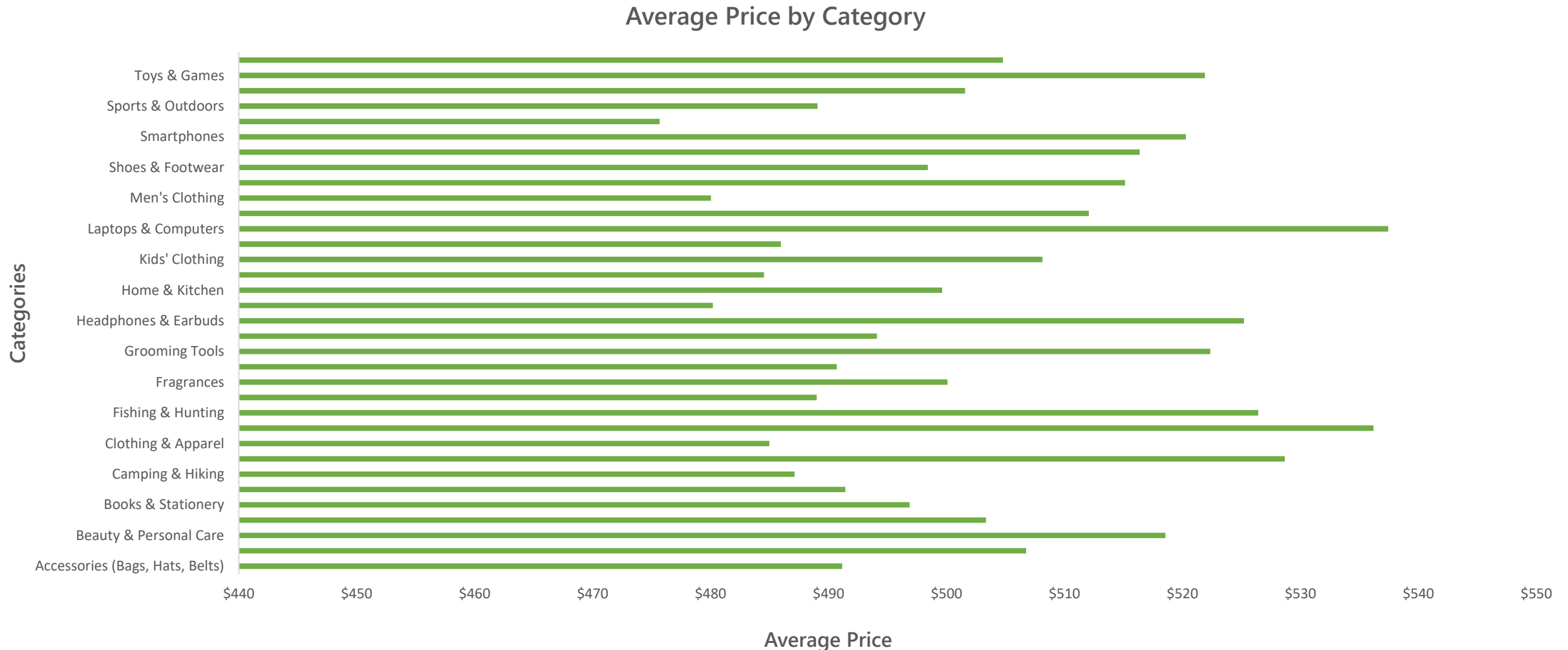
Q2. What is the availability statuses of product distribution?

The availability of product is fairly balanced across all statuses, with discontinued and out of stock appearing slightly more often than the others.



Q3. How do the average product price vary across categories?

Average prices across categories show minimal variation, ranging from \$476 - \$537.



Key Insights & Limitations

- **Insights:**

- Product distribution is fairly balanced across categories, with no strong dominance.
- Product availability is evenly spread, with discontinued and out of stock products slightly more common.
- Pricing is clustered across categories(\$476 - \$537), showing minimal variations.
- Overall, the dataset is stable and well distributed.

- **Limitations:**

- No date/time column, limiting basic trend analysis.
- Brand column required grouping due to repetitive and inconsistent naming
- Prices are very close across categories, so it is hard to clearly see the variation between them.